

Nicolas Villalobos

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EDUCATION

University of Kansas

Lawrence, KS

B.S. in Electrical Engineering & Computer Science; GPA: 3.53

May 2029

Coursework: *Digital Logic Design, Programming 2 (Intro to Data Structures and Algorithms), Multivariable Calculus, Probability and Statistics for Computer Scientists, Software Engineering I, Discrete Structures, and Linear Algebra.*

Organizations: *Institute of Electrical and Electronics Engineers (IEEE), Association for Computing Machinery (ACM), Blockchain Institute and Society of Hispanic Professional Engineers (SHPE).*

EXPERIENCE

University of Kansas, Department of EECS

Lawrence, KS

Undergraduate Research Assistant, Computer Vision

Aug 2026 – Present

- Implement experiment code in Python for a graduate thesis on **privacy-preserving face membership verification**, running parameter sweeps over Bloom filter size, hash count, and hierarchy depth to generate the membership-testing results used in the thesis evaluation.
- Built the **evaluation harness** comparing region-based Hierarchical Bloom Filter layouts against the uniform-hierarchy baseline across **five face datasets**, handling dataset preprocessing, dlib landmark extraction, and binary template generation under matched storage budgets.
- Maintain the project codebase, result logs, and reproducibility scripts, and present implementation status in weekly meetings with faculty and the thesis author.

PROJECTS

NanoGPT Language (Bribri) Model – Python, PyTorch, BPE Tokenization, Gradio

Apr 2026 – Jul 2026

- Built a **788K-parameter GPT from scratch** in PyTorch (RoPE, RMSNorm, SwiGLU) with no HuggingFace Trainer, achieving 32.1 validation perplexity in 2,000 steps trained on CPU via cosine LR scheduling and gradient clipping.
- Assembled Bribri indigenous language corpus by collecting data across 4 academic sources via mojibake encoding repairing, producing **13,327 unique records and 208K BPE tokens** for a language with only ~7,000 living speakers.
- Trained a custom 1,024-vocab BPE tokenizer optimized for Bribri's writing system that **reduced tokenization overhead by 74% vs GPT-2** (2.1 vs 3.65 tokens/word), cutting total sequence length from 355K to 205K tokens.

SWE-bench Coding Agent – Python, Gemini (API), Docker

Apr 2026 – Jul 2026

- Engineered **autonomous coding agent from scratch** (no frameworks). Resolves GitHub issues end-to-end, via tool-use agentic loop with 5 custom tools, unique-match code editing, token **budget tracking**, and per-task Docker sandboxing.
- Produced **valid patches for 43% of SWE-bench Lite tasks**, generating 500+ character differences across a 30-task benchmark, with structured JSONL trajectory logging and an HTML replay viewer for debugging.
- Built a **3-stage scaffolding pipeline** (file localization, fix planning, self-verification) and ablation framework comparing 4 configurations across a 10-category failure taxonomy with reward-hacking detection, at **\$0 cost via free-tier rotation**.

Structure-from-Motion 3D Reconstruction Pipeline – Python, OpenCV, NumPy, SciPy

Jul 2026 - Aug 2026

- Built an incremental Structure-from-Motion pipeline using SIFT matching, RANSAC pose estimation, and triangulation, registering 47/47 cameras into a **21,020-point reconstruction** with 0.165px reprojection error and **6.71° rotation error**.
- Debugged two silent correctness bugs using ground-truth pose comparisons, fixing a mislabeled reference camera and a triangulation defect that produced outliers 1,377 units from a 5-unit point cluster, removing 487 stray points.
- Optimized a **65,000-parameter** bundle adjustment using a sparse Jacobian over **43,014 observations**, cutting **reprojection error 13.6%** (0.191px to 0.165px) in 90 seconds after resolving a robust-loss solver stall.

Wildfire Burn Severity Mapping – Python, Google Earth Engine, geemap

Jul 2026 – Aug 2026

- Queried and cloud-masked 232 Sentinel-2 surface-reflectance scenes, filtering to 105 pre-fire and 54 post-fire images via a 7-class SCL mask, compositing imagery across a 1,076,178-hectare AOI at 20-meter resolution before differencing.
- Computed dNBR and applied a Key & Benson offset correction for a measured -0.1016 median seasonal shift, moving burned area from 255,260 ha to 337,823 ha and closing the gap to the official 389,837 ha fire size from **-34.5% to -13.3%**.
- Reran classification without SCL cloud/snow masking: masking raised detected High-severity area from **41,084 ha to 64,080 ha**, showing contamination suppressed **35.9% of real burn** signal rather than manufacturing false positives.

TECHNICAL SKILLS

- **Languages:** Python, Swift, JavaScript, HTML/CSS, C/C++
- **AI & ML:** PyTorch, Hugging Face Transformers, dlib, BPE tokenization, Gradio, NumPy, SciPy, OpenCV
- **Tools & Platforms:** Git/GitHub, Docker, pytest, Jupyter, Xcode, Supabase